

Linear Algebra and Several Variable Calculus Notes

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Abstract

This is a simple abstract for the notes.

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20% participation/workshop, 80% final exam.

1 Lecture 1

- solve Linear equations
- define vectors/ dot product
- systems of equations
- matrixs

1.1 work with real numbers

introduce $\mathbb{R}^n$ (actually a vector space)

1.1.1 nature of $\mathbb{R}$

if $xy \in \mathbb{R}$, then

$$\begin{aligned}xy &= yx \quad (\text{for matrices this may not be true}) \\x + y &= y + x \\-x &\in \mathbb{R} \\x^{-1} &\in \mathbb{R} \quad (x \neq 0)\end{aligned}$$

1.1.2 define points in space

$$\begin{aligned}\dim = 2 &: \mathbb{R}^2, (x, y) \\ \dim = 3 &: \mathbb{R}^3, (x, y, z) \\ \dim = n &: \mathbb{R}^n, (x_1, x_2, \dots, x_n)\end{aligned}$$

1.1.3 addition and subtraction in $\mathbb{R}^n$

in $\mathbb{R}^n$ we can add and subtract vectors
for example

$$A = (a_1, a_2, \dots, a_n), B = (b_1, b_2, \dots, b_n) \\ A + B = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

1.1.4 scalar multiplication

Let c be a scalar in $\mathbb{R}$. The definition is

$$cA = (ca_1, ca_2, \dots, ca_n).$$

1.2 vectors

vectors in $\mathbb{R}^3$ is Ordered pairs "pairs"

vector: from A point to B point, we can easily move them around

$$\overrightarrow{AB} = B - A$$

$\overrightarrow{AB}$ and $\overrightarrow{CD}$ are parallel if

$$\overrightarrow{AB} = k\overrightarrow{CD}$$

1.2.1 representation of vectors

We have two way to represent vectors.

row vector: $[x_1, x_2, \dots, x_n]$ vectors in row space

column vector: $[x_1, x_2, \dots, x_n]^T$ vectors in column space

to understand how the rows and columns are related to each other, we have fundamentum theorem of linear algebra.

row space and column space have the same dimension

so row rank are equals to column rank.

1.2.2 define dot product

Dot product is a useful tool to describe different relations related to parallelism
Assume

$$A = (a_1, a_2, \dots, a_n), \quad B = (b_1, b_2, \dots, b_n) \\ A \cdot B = a_1b_1 + a_2b_2 + \dots + a_nb_n$$

A and B are vectors in $\mathbb{R}^n$

length: $\|A\| = \sqrt{A \cdot A} = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$ is a real number

perpendicular: $A \cdot B = 0$

For understanding dot product, consider A,B. First project A to B at point P, c is a scalar.

$$\overrightarrow{P} = c\overrightarrow{B}$$

And

$$\overrightarrow{AP} = A - cB$$

Then because $\overrightarrow{AP}$ is perpendicular to B .

$$\begin{aligned}(A - cB) \cdot B &= 0 \\ A \cdot B - cB \cdot B &= 0 \\ A \cdot B - c\|B\|^2 &= 0 \\ c &= \frac{A \cdot B}{\|B\|^2}\end{aligned}$$

Dot product is a projection of A onto B resulting in a scalar c .

And we have $\cos(\theta)$

$$\cos(\theta) = \frac{c\|B\|}{\|A\|}$$

So combining the equations gives

$$A \cdot B = \|A\| \|B\| \cos(\theta)$$

The meaning of dot product is calculate how much of A parallel to B And think what you are working with: scalar or vector.

1.3 lines and planes

1.3.1 consider lines

for a vector $X = P + tA$, where P and A are vectors in $\mathbb{R}^n$ and t is a scalar
Consider a line in $\mathbb{R}^2$. For

$$X = (x, y) = (p, q) + t(a, b),$$

we obtain two equations:

$$\begin{aligned}x &= p + ta \\ y &= q + tb\end{aligned}$$

Change the first and the second equation.

$$\begin{aligned}bx &= bp + tab \\ ay &= aq + tab\end{aligned}$$

subtraction two equations.

$$bx - ay = bp - aq$$

times $\frac{1}{a}$

$$\begin{aligned}\frac{b}{a}x - y &= \frac{bp - aq}{a} \\ y &= \frac{b}{a}x - \frac{bp - aq}{a} \\ y &= mx + c\end{aligned}$$

is the non-vector equation of the line.

1.3.2 planes

Consider a plane in $\mathbb{R}^3$, defined by a vector N perpendicular to the plane and a point P in the plane.

Assume $X = (x, y, z)$, $N = (a, b, c)$

$$\begin{aligned}(X - P) \cdot N &= 0 && \text{(perpendicular)} \\ ax + by + cz - N \cdot P &= 0 && N \cdot P \text{ is a scalar} \\ ax + by + cz &= d\end{aligned}$$

1.4 matrices

definition of matrix originally from equations.

$$\begin{aligned}2x - y &= 0 \\ -x + 2y &= 3\end{aligned}$$

And for matrix

$$\begin{aligned}A &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 3 \end{bmatrix} \\ \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ 3 \end{bmatrix} \\ AX &= B\end{aligned}$$

2 Lecture 2

- Matrix (dot/row/column)
- Transpose (swap rows+ columns), index rotation
- Eliminations(next lecture)

Elimination: relative to machine learning, make solving linear systems more efficient.

2.1 Equations

$$\begin{aligned}2x - y &= 0 \\ -x + 2y &= 3\end{aligned}$$

matrix form

$$\begin{aligned}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \\ \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ 3 \end{bmatrix}\end{aligned}$$

$AX=B$,

A matrix, X column vector, B column vector.

2.2 Represent

Equations by matrices. let n, m be 2 positive integers.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \ddots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix},$$

A_{ij} : first index corresponds to the row, and second index corresponds to the column.

2.2.1 Zero matrix

Zero matrix: all elements are zero.

$$0_{n \times m} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

2.2.2 Identity matrix

Identity matrix: square matrix with ones on the diagonal and zeros elsewhere.

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

2.3 Addition and scalar multiplication

2.3.1 Addition matrices

$$A = \begin{bmatrix} -2 & 1 & 0 \\ 2 & 3 & 4 \end{bmatrix}_{2 \times 3}, \quad B = \begin{bmatrix} 5 & 1 & -1 \\ 2 & 1 & -1 \end{bmatrix}_{2 \times 3}$$

then

$$A + B = \begin{bmatrix} -2 + 5 & 1 + 1 & 0 + (-1) \\ 2 + 2 & 3 + 1 & 4 + (-1) \end{bmatrix} = \begin{bmatrix} 3 & 2 & -1 \\ 4 & 4 & 3 \end{bmatrix}$$

addition need to be same dimension/size.

2.3.2 Scalar multiplication

$$A = \begin{bmatrix} -2 & 1 & 0 \\ 2 & 3 & 4 \end{bmatrix}, \quad k = 2$$

then

$$kA = 2 \begin{bmatrix} -2 & 1 & 0 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} -4 & 2 & 0 \\ 4 & 6 & 8 \end{bmatrix}$$

2.4 Transpose

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 5 \end{bmatrix}_{2 \times 3}, \quad A^T = \begin{bmatrix} 2 & 1 \\ 1 & 3 \\ 0 & 5 \end{bmatrix}_{3 \times 2}$$
$$B = [1 \quad 5 \quad 7]_{1 \times 3}, \quad B^T = \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}_{3 \times 1}$$

2.5 Matrix multiplication

- Dot-way
- Column-way
- Row-way

2.5.1 Dot way

$$\begin{aligned} 2x - y &= 0 \\ -x + 2y &= 3 \end{aligned}$$
$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

Row multiply by the column.

Row vector dot the column vector.

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

weight of each row vector in the linear combination.

2.5.2 Column way

Note $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$ is combination of column vectors $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$.

$$\begin{aligned} [a_{11} \quad a_{12} \quad a_{13}]_{1 \times 3} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}_{3 \times 2} &= [a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} \quad a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32}] \\ &= [c_{11} \quad c_{12}] \end{aligned}$$

2.5.3 Row way

$$\begin{cases} x + y = 0 \\ 2x + 2y = 4 \rightarrow x + y = 2 \end{cases}$$

$$\begin{cases} x + y = 0 \\ x + y = 2 \end{cases}$$

Can write by matrix form

$$x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad \text{which has no solution}$$

And consider

$$\begin{bmatrix} 1 & 3 \\ 8 & 4 \end{bmatrix}$$

try

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 8 & 4 \end{bmatrix} = \begin{bmatrix} 8 & 4 \\ 1 & 3 \end{bmatrix}$$

swaps the two rows.

2.6 Index rotation

2.6.1 index rotation

A_{ij} : i is the row and j is the column. transpose

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 7 \\ 2 & 0 \end{bmatrix}_{3 \times 2}, \quad A^T = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 7 & 0 \end{bmatrix}_{2 \times 3}$$

$$a_{11} = 1$$

$$a_{21} = 3$$

...

Set

$$B = A^T$$

Elements of B_{ij} is in term of A_{ij} .

Specifically $b_{ij} = a_{ji}$ is the operation that swaps the rows and the columns.

2.6.2 $(AB)^T = B^T A^T$

Set two matrixs A and B, claim that

$$(AB)^T = B^T A^T$$

let a_{ij} be the element of matrix A and b_{jk} be the element of matrix B.

$$C = AB$$

then

$$c_{ij} = \sum_k a_{ik} b_{kj} \quad \text{dot-way}$$

Let b'_{kj} be the element of matrix B^T and let a'_{ji} be the element of matrix A^T .

$$D = B^T A^T$$

Then the element of $(B^T A^T)_{ij}$ is

$$d_{ki} = \sum_j b'_{kj} a'_{ji} = \sum_j b_{jk} a_{ij} = \sum_j a_{ij} b_{jk} = c_{ik}$$

Show that

$$(B^T A^T)_{ki} = c_{ik} = (C^T)_{ik}$$

Therefore,

$$(AB)^T = B^T A^T$$

3 Lecture 3

- Geometry of linear equations
- Elimination

3.1 Transpose

Swap rows and columns

The row become the column and the column become the row.

Every element a_{ij} have two index, the first one is the row and the second one is the column. It's always the row index first, then the column index.

Transpose exchange row and column number.

$$(m \times n)^T = n \times m$$

Also emphasize that the transpose of a product of matrices follows the rules, we prove that by index rotation last lecture.

$$(AB)^T = B^T A^T$$

is really important for quantum mechanics.

3.2 Division

It actually means the inverse, only square matrix has inverse.

$$AA^{-1} = I$$

(here A is a square matrix and A^{-1} is its inverse.)

3.3 identity

I is identity matrix, B is any matrix that can be multiplied by I .

$$IB = B$$
$$I = \begin{bmatrix} 1 & 0 & \cdots \\ 0 & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

3.4 Inverse of transpose

$$\text{Claim: } (A^T)^{-1} = (A^{-1})^T$$

We start with the identity

$$\begin{aligned}(AA^{-1})^T &= I^T \\ (A^{-1})^T A^T &= I \\ ((A^T)^{-1}) A^T &= I\end{aligned}$$

Therefore

$$(A^T)^{-1} = (A^{-1})^T$$

”nitty property of transpose and inverse.”

3.5 Equations

Geometry

3.5.1 Case 1

$$\begin{aligned}x + y &= 0 \\ 2x + 2y &= 5\end{aligned}$$

matrix way

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

Column way

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

from geometry scalar multiply of $(1, 2)$ can not form the vector $(0, 5)$, ∞ solution.

3.5.2 Case 2

$$\begin{aligned}2x - y &= 0 \\ -x + 2y &= 3\end{aligned}$$

matrix way

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

Column way

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

two vector not the same vector so can form the vector $(0, 3)$, solution $1\text{col}1 + 2\text{col}2 = (0, 3)$.

3.5.3 Case 3

$$\begin{aligned}5x + y &= 6 \\10x + 2y &= 12\end{aligned}$$

matrix way

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \end{bmatrix}$$

Column way

$$\begin{aligned}x \begin{bmatrix} 5 \\ 10 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \end{bmatrix} &= \begin{bmatrix} 6 \\ 12 \end{bmatrix} \\5x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \end{bmatrix} &= \begin{bmatrix} 6 \\ 12 \end{bmatrix}\end{aligned}$$

two vector are parallel, hence the system has infinitely many solutions same as case1.

3.6 Homogeneous equations

$$A\mathbf{x} = \mathbf{0}$$

A matrix x column vector of variables equals the zero vector.

3.6.1 Elimination

$$\begin{aligned}2x - y &= 0 \\3x + y &= 0\end{aligned}$$

matrix way

$$\begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or dot multiplication

$$\begin{aligned}(2, -1) \cdot \begin{bmatrix} x \\ y \end{bmatrix} &= 0 \\(3, 1) \cdot \begin{bmatrix} x \\ y \end{bmatrix} &= 0\end{aligned}$$

$A_1 = (2, -1)$, $A_2 = (3, 1)$, $X = (x, y)$, c is a real number.

$$\begin{aligned}A_1 \cdot X &= 0 \\A_2 \cdot X &= 0 \\(A_1 + A_2) \cdot X &= 0 \\cA_1 \cdot X &= 0\end{aligned}$$

Elementary row operations

- Add rows
- permute rows
- multiply a row by a nonzero scalar

3.6.2 Example

$$\begin{aligned}2x - y &= 0 \\ -x + 2y &= 3\end{aligned}$$

matrix way

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$
$$AX = B$$

Apply row operations to solve the system

Augmented matrix

$$\begin{aligned}& \left[\begin{array}{cc|c} 2 & -1 & 0 \\ -1 & 2 & 3 \end{array} \right] \Rightarrow R_2 = 2R_2 + R_1 \\& \sim \left[\begin{array}{cc|c} 2 & -1 & 0 \\ 0 & 3 & 6 \end{array} \right] \Rightarrow R_2 = R_2/3 \\& \sim \left[\begin{array}{cc|c} 2 & -1 & 0 \\ 0 & 1 & 2 \end{array} \right] \Rightarrow R_1 = R_1 + R_2 \\& \sim \left[\begin{array}{cc|c} 2 & 0 & 2 \\ 0 & 1 & 2 \end{array} \right] \Rightarrow R_1 = R_1/2 \\& \sim \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array} \right]\end{aligned}$$

Hence, the solution is:

$$x = 1, \quad y = 2$$

3.6.3 3rd order system

$$\begin{aligned}x + 2y + z &= 2 \\ 3x + 8y + z &= 12 \\ 4y + z &= 2\end{aligned}$$

matrix way

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix}$$

augmented matrix

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array} \right] \Rightarrow R_2 = R_2 - 3R_1 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 4 & 1 & 2 \end{array} \right] \Rightarrow R_3 = R_3 - 2R_2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array} \right] \Rightarrow R_3 = R_3/5, R_2 = R_2/2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \end{array} \right] \Rightarrow R_2 = R_2 + R_3, R_1 = R_1 - R_3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array} \right] \Rightarrow R_1 = R_1 - 2R_2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array} \right]
 \end{aligned}$$

Hence, the solution is:

$$x = 2, \quad y = 1, \quad z = -2$$

There are two ways to get solutions, keep on going like that, or end at upper triangular form and then perform back substitution.

Upper triangular form.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \end{array} \right]$$

RREF Row reduced form

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array} \right]$$